

NanoGT Match Report: Pompe disease

Disease: Pompe disease (ORPHA:365)

Primary gene: GAA

Gene CDS: 2856 bp

Inheritance: Autosomal recessive

Target tissues: muscle, heart

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	8.4/10	[GREEN] High	approved
2	ABO-101	AAV9	7.9/10	[GREEN] High	phase1/2
3	AVR-RD-01	LV	7.9/10	[GREEN] High	phase1/2
4	RGX-121	AAV9	7.9/10	[GREEN] High	phase3
5	ST-920	AAV2/6	7.8/10	[GREEN] High	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 8.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	8.39	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2856bp / cargo 8000bp (36% utilized)
- No tissue overlap (disease: ['muscle', 'heart'], vector: ['hematopoietic'])
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #2: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	2.00	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.94	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlap: heart, muscle
- Both lysosomal proteins — cross-correction likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #3: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.94	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2856bp / cargo 8000bp (36% utilized)
- No tissue overlap (disease: ['muscle', 'heart'], vector: ['hematopoietic'])
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient

- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #4: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 7.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.94	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlap: heart, muscle
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage

- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #5: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 7.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
TOTAL (normalised)	7.78	10.0	Raw sum / 18×10

Rationale

- Gene CDS 2856bp / cargo 4700bp (61% utilized)
- Vector tropism overlaps muscle, but precedent target is liver
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera